UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA
Faculty of Engineering, Built Environment and Information Technology

School of Information Technology
Department of Computer Science
COS326 – Database Systems
Practical 3 2026

Release Date:	15 September 2026
Submission Date and Time:	29 September 2026 @ 23:59Hrs
Lecturer:	Mr S.M Makura
Total: 50 Marks	

A. Objectives

1. Apply XPath expressions to retrieve and filter information from XML documents.
2. Construct XQuery FLWOR expressions to query and analyse XML data stored in BaseX.
3. Perform cross-document queries by combining related information from multiple XML documents.
4. Use XQuery functions such as count(), sum(), avg(), max(), min(), and distinct-values() to perform data analysis and aggregation.
5. Apply filtering, ordering, grouping, and conditional logic to produce meaningful XML-based analytical reports.
6. Develop XQuery solutions that demonstrate an understanding of XML structure, relationships between documents, sequences, predicates, and FLWOR expressions.

B. Submission Procedure

1. You must submit a filed named Prac3Queries.txt which contains all your queries.
2. The completed **AI declaration form** where you detail the AI prompts you used and the usage types.
3. Compress the above documents into an archive and upload it to the ClickUP link for practical submissions before the due date/time. The file name for the archive must have your student number as part of the file name, e.g.

uXXXXXXXX_Surname_Initials(XXXXXXXX is your student number) e.g u12345678_Smith_JN

NO LATE submissions will be accepted after the submission date and time has lapsed. Do not wait until the last minute to submit and then provide reasons related to technical difficulties.

Important

- All queries must be written and tested using BaseX.
- Your queries must operate on the supplied XML documents and must not rely on manually entered results.

Academic Integrity: You may use AI-assisted tools (e.g., ChatGPT, Copilot, Gemini etc) to help you understand XQuery concepts and syntax. However, you remain responsible for the design, implementation, and correctness of your solution. During marking, you may be asked to explain or modify any part of your submission. Failure to demonstrate understanding may result in marks being deducted.

Question 1: Wildlife Conservation Investigation

Scenario

You have been appointed as a junior data analyst for a wildlife conservation organisation responsible for monitoring endangered and vulnerable animals across several South African wildlife reserves.

The organisation maintains information about individual animals in one XML document and field observations in a separate XML document.

The supplied XML database consists of:

- **wildlife.xml** - contains information about individual animals, including their species, age, conservation status, and home reserve.
- **observations.xml** - contains records of field observations, including the animal observed, reserve, observation date, duration, and number of individuals recorded.

You will use BaseX and XQuery to investigate and analyse this data.

Source: Makura S.M (2026)

Task 1: XPath Investigation [5 Marks]

Using XPath only, retrieve the names of all animals that are older than 10 years and whose conservation status is not Least Concern.

Your query must:

- use an appropriate XPath expression;
- correctly filter animals based on both conditions; and

Task 2: Cross-Document Analysis [8 Marks]

The conservation organisation wants to identify animals that have been observed outside their registered home reserve**.**

Using a FLWOR expression**,** query information from both wildlife.xml and observations.xml.

For every animal that has been observed in a reserve different from its home reserve, return:

- animal name;
- home reserve; and

- observed reserve.

Each animal/reserve combination must appear only once**,** even if the animal was observed there multiple times.

The results must be ordered **alphabetically by animal name** and then by observed reserve.

Return the results using the following XML structure:

```
<movement>
    <animal>...</animal>
    <homeReserve>...</homeReserve>
    <observedReserve>...</observedReserve>
</movement>
```

Your query must demonstrate:

- a relationship between the two XML documents;
- appropriate use of for / let;
- filtering using where;
- elimination of duplicate values; and

Task 3: Reserve Aggregation Analysis [8 Marks]

The organisation wants to determine which reserves account for an unusually large number of observed individuals.

For each reserve**,** calculate the total number of individuals recorded across all observations**.**

Then calculate the average total number of observed individuals across all reserves**.**

Return only those reserves whose total is greater than the average reserve total.

The results must be ordered from the **highest total to the lowest total**.

Return the results using:

```
<reserveSummary>
    <reserve>...</reserve>
    <totalObserved>...</totalObserved>
</reserveSummary>
```

Your query must demonstrate the use of:

- `distinct-values();`
- `sum();`
- `avg();`
- FLWOR expressions;
- filtering based on an aggregate value; and
- descending ordering.

Task 4: Species-Level Analysis [9 Marks]

The conservation organisation wants to compare observation activity across different animal species.

For each species, determine:

1. the total number of observations recorded for that species; and
2. the average number of individuals recorded per observation.

Only include species with **at least two observations**.

The average group size must be displayed to two decimal places**.**

The results must be ordered from the highest average group size to the lowest.

Return the results using:

```
<speciesAnalysis>
    <species>...</species>
    <observations>...</observations>
    <averageGroupSize>...</averageGroupSize>
</speciesAnalysis>
```

Your query must correctly establish the relationship between:

- an animal's species in wildlife.xml; and
- its observations in observations.xml.

Your solution should demonstrate the use of:

- `distinct-values();`
- `count();`
- `avg();`
- filtering;
- ordering; and
- appropriate numeric formatting/rounding.

Task 5: Most Frequently Observed Animal [7 Marks]

The conservation organisation wants to determine which individual animal has been monitored most frequently.

For **each animal**, calculate the number of observations recorded for that animal.

Identify the animal or animals with the **highest number of observations**.

Your query must correctly handle the possibility of a **tie**. Therefore, your solution must not assume that there will always be only one most frequently observed animal.

Return the results using:

```
<mostObserved>
    <animal>...</animal>
    <observationCount>...</observationCount>
</mostObserved>
```

If two or more animals have the same maximum number of observations, all qualifying animals must be returned.

The results must be ordered alphabetically by animal name.

Your solution must demonstrate:

- counting observations for each animal;
- determining the maximum observation count;
- filtering based on that maximum; and
- handling ties correctly.

Task 6: Conservation Risk Report [7 Marks]

The organisation is concerned that some animals with elevated conservation risk may not be receiving sufficient monitoring.

Determine the **average number of observations per monitored animal** across the entire database.

Then identify animals that satisfy **both** of the following conditions:

- their conservation status is either **Vulnerable** or **Endangered**; and

- their individual observation count is below the average number of observations per monitored animal**.**

For each qualifying animal, return:

- animal name;
- conservation status; and
- number of observations.

Use the following structure:

```
<monitoringAlert>
    <animal>...</animal>
    <status>...</status>
    <observationCount>...</observationCount>
</monitoringAlert>
```

Order the results by conservation status and then by animal name.

Your query must demonstrate:

- calculation of an overall average;
- counting observations for individual animals;
- multiple Boolean conditions;
- comparison against an aggregate value; and
- appropriate ordering.

Task 7: Animal Range Analysis [6 Marks]

The conservation organisation wants to investigate animals that have been observed across multiple reserves.

For every animal that has been observed in **more than one distinct reserve**, determine:

1. the animal's name;
2. its registered home reserve;
3. the number of distinct reserves in which it has been observed; and
4. the reserve or reserves where it has been observed **most frequently**.

Your query must correctly handle **ties**. If an animal has been observed the same number of times in two or more reserves, all tied reserves must be reported.

Return the results using:

```
<rangeAnalysis>
    <animal>...</animal>
    <homeReserve>...</homeReserve>
    <reserveCount>...</reserveCount>
    <mostFrequentReserves>
        <reserve>...</reserve>
    </mostFrequentReserves>
</rangeAnalysis>
```

The results must be ordered alphabetically by animal name.

Your solution must demonstrate:

- use of distinct-values();
- counting observations by reserve;
- identifying a maximum value;
- handling ties;
- nested FLWOR expressions; and
- construction of structured XML output.

Mark allocation breakdown:

Task	Description	Marks
Task 1	XPath Investigation	**5**
Task 2	Cross-Document Analysis	**8**
Task 3	Reserve Aggregation Analysis	**8**
Task 4	Species-Level Analysis	**9**
Task 5	Most Frequently Observed Animal	**7**
Task 6	Conservation Risk Report	**7**
Task 7	Animal Range Analysis	**6**
	TOTAL	**50**